Update, 2026-11-11. Under-prediction is model-dependent (float64: CHGNet 25/28, MACE-MP-0 medium 21/29, MACE-MP-0 small 17/26, MACE-MPA-0 medium 13/28). The physical pilot's preregistered gate failed; its passing check is same-engine only. See "Shared DFT Anchors" (lupine.science/articles/shared-dft-anchors/). The July 2026 text below is otherwise unrevised.

A Correction Boundary for Foundation Interatomic Potentials: Machine-Checked Limits, a Geometry–Levels Decomposition, and Model-Guided Sparse DFT

Memorandum of theory, draft v0.2 — 2026-07-20.

> **Frozen-before-results declaration.** Every sentence of framing, theory, and protocol was written before the sparse-DFT campaign completed. Result slots are marked `[PENDING]`; everything measured is locked `[FINAL]` with receipts. Claims carry an epistemic register (§9). Machine-checked theorems: `lean-spec/OpenDistillationFactory/HonestErrors/CorrectionBoundary.lean` — merged, 3761-job `lake build`, zero `sorry`, zero new axioms (`#print axioms` restricted to `propext`, `Classical.choice`, `Quot.sound`). Preregistrations and locked panels are content-addressed; the abstention record is included, not edited away.

Intended readership. Machine-learning venues (ICML track on science), computational-materials venues, and reviewers who must decide whether AI-designed matter can be trusted — including those who audit claims on behalf of institutions and states.

Abstract

Foundation machine-learned interatomic potentials (uMLIPs) are being asked to carry materials decisions at national scale — battery electrolytes, catalysts, structural alloys — on the strength of speed claims that were never falsification-tested at the observables that matter. We report the first preregistered, hash-locked measurement of that trust question on migration barriers: across 30 chemistry-held-out lithium-conductor paths, four foundation models miss a 40 meV accuracy gate by 3.4–6× (MAE 135.0–242.5 meV), identically under two precision regimes, with systematic under-prediction rather than scatter `[FINAL]`. We then do three things the field has not done. First, we prove the *correction boundary* in Lean 4: additive shifts cannot move a barrier; a model's barrier error is exactly its profile wobble, with a tightness witness; slope corrections help only under sign stability; and anchors at the two true extrema reconstruct the barrier exactly — which is to say, accuracy costs precisely model irrelevance at two points (§3). Second, we show the failure decomposes: the models' energy *levels* are wrong while their energy *landscape geometry* — transition-state locations — is right on 82–86% of paths (89–93% within one image) `[FINAL]`. Third, from the theorems and the decomposition we derive **model-guided sparse DFT**: the model proposes

extrema, a free DFT engine measures only those points. On the locked reference profile the protocol reproduces barriers with MAE 1.2–9.4 meV at ≈7 evaluations per path, against ≈50–100 for full NEB — under the same 40 meV gate the raw models miss by 3.4–6× `[FINAL, simulation]`. The physical campaign is frozen herein (protocol, success criteria, analysis plan; §5) and reports `[PENDING]`. Beyond the specific result, we present the process as a method for distilling physical laws from error structure: preregistration with content-addressed evidence, anti-laundering ingestion into machine-checked claims registries, and abstention as a first-class epistemic output (§7). The claims that could not survive this process — every cheap correction we tried — are preserved as a record.

1. Introduction

1.1 The trust question, asked properly

The public case for foundation interatomic potentials is throughput: screen thousands of candidates at MLIP speed, let DFT follow the winners. This case fails silently if the MLIP errs *systematically* — if every barrier it reports is low and every interface it reports is weak — because then the winners are chosen by a biased judge, and no amount of downstream DFT launders the choice. The correct response to a biased judge is not more screening; it is a boundary: a formal statement of where the judge can and cannot be believed, and a protocol that operates inside the boundary.

We build that boundary here, in three layers: measured failure (§2), machine-checked limits (§3), and a protocol that passes inside the limits (§4–§5). Each layer is constrained to be checkable by an adversarial reviewer: measurements are hash-locked before analysis, theorems are machine-checked, and anything not yet measured is explicitly marked pending.

1.2 Contributions

1. **A preregistered measurement of systematic underbinding** at migration barriers across 30 chemistry-held-out paths and four foundation models, in duplicate precision regimes, with all artifacts content-addressed (§2).
2. **The correction boundary**: to our knowledge the first machine-checked impossibility bounds for runtime correction of uMLIP observables — shift invariance, the wobble bound with tightness, slope-stability licensing, and anchor pricing with two-point exactness (§3).
3. **The geometry–levels decomposition**: the empirical separation of energy-level error from landscape-geometry accuracy, quantified per model (§4).
4. **Model-guided sparse DFT**, a protocol that passes the same gate the raw models fail, at an order-of-magnitude-lower DFT cost, with its cost and failure modes declared in advance (§5).
5. **The process itself** as a method for distilling physical laws from error structure: preregistration, anti-laundering evidence chains, machine-checked claims, and abstention-as-output (§7).

2. The measured failure

2.1 Protocol

Thirty migration paths, one per chemical system, drawn from the LiTraj nebDFT2k benchmark (npj Comput. Mater. 2025, DOI 10.1038/s41524-025-01571-z) and locked with SHA-256 `192fe54a…` before any model ran. The barrier convention — max(image energy) − min(image energy) — is identical on the reference and prediction sides; the panel builder validates the dataset's own barrier against the profile max−min within 0.5 meV. Execution: one Cloud Run cell per model on isolated jobs (NVIDIA L4), CI-NEB per path under a frozen protocol, per-path outcomes recorded without imputation; manifest and panel hashes validated fail-closed by the runner.

2.2 Results `[FINAL]`

Model	float32 MAE (meV)	float64 MAE (meV)	completed	signed-error direction
mace-mpa-0-medium	135.0	135.0	28/30	17/26 negative (mp-small f64)
mace-mp-small	152.0	151.9	26/30	all 26 negative (f32)
mace-mp-medium	174.7	174.7	29/30	mixed
chgnet	242.5	242.5	28/30	predominantly negative

A float32 protocol defect found in review (vendor guidance reserves float32 for geometry optimization) changed the outcome by ≤0.1 meV per model: the failure is model error, not numerics. This replicates, at preregistered scale, the published softening record (Deng et al., DOI 10.1038/s41524-024-01500-6; Bheemaguli et al., DOI 10.1039/D5DD00534E, who report 73–78% underprediction across 574 literature-derived paths).

2.3 Every cheap correction fails — and the record is kept

Global scalars, additive shifts, per-family shifts, and frozen six-point deltas leave MAE at 139–160 meV or worse. A preregistered disjoint-training pilot (12 paths, zero chemistry overlap) ended in universal abstention: signed errors flip sign *within* every chemistry family, so no low-dimensional correction is honest on this field. Even a perfect best-of-four oracle — hindsight model selection per path — reaches only 70 meV. We preserve these results because they are the evidence that the boundary in §3 is not optional.

3. Theory: the correction boundary

Definitions. A profile is a map `p : Fin n → ℝ` (sampled energies along a reaction path); `barrier p = max p − min p`; the per-image deviation `d = m − r` between model and reference profiles; the wobble

`wobble d = max d − min d`. All theorems below are proved in Lean 4 (`CorrectionBoundary.lean`) with zero `sorry`.

Theorem 3.1 (shift invariance). For any constant `c`, `barrier (fun i ↦ p i + c) = barrier p`. *Global additive shifts cannot change a barrier.* This kills the entire family of level-shift corrections — not with an experiment, with a theorem.

Theorem 3.2 (wobble bound). `|barrier m − barrier r| ≤ wobble d`, and there exist profiles achieving equality (tightness witness `barrier_error_le_wobble_tight`). A model's barrier error is *exactly* its profile wobble; no correction operating on the model's own outputs can guarantee better. This is the no-free-lunch theorem of runtime correction.

Theorem 3.3 (slope family). For a single-ratio correction scaled by `σ`, the corrected error is bounded by the slope residuals (`slope_correction_error_le`); one-signed slope ratios are exactly the condition under which the midrange-ratio certificate strictly improves the raw one; and there exist mixed-sign profiles on which every ratio correction degenerates to a shift — whereupon Theorem 3.1 applies (`slope_instability_witness`). Note carefully what this says: slope corrections are *licensed only by sign stability*, a condition our training data violated everywhere (§2.3) — the abstention is a theorem's worth of honesty, not a conservative mood.

Theorem 3.4 (anchor pricing). Anchoring a single point `k` to its reference value changes the barrier by at most `|m k − r k|` (`anchor_barrier_change_le`), and the residual error is at least the raw error minus that amount (`anchor_impossibility_bound`). Anchors at the sampled maximum and minimum reconstruct the barrier exactly (`two_point_anchor_exact`). Read as a price list: accuracy costs exactly two anchors — and at two anchors the model contributes nothing but location. (A stronger "precise" bound proposed in the design was found false and is refuted in-file, `anchor_wobble_lower_bound_refuted`: the boundary layer audits itself.)

Corollary 3.5 (routing bound). Any model-selection rule restricted to a fixed model set inherits each member's wobble; hence the 70 meV oracle ceiling of §2.3 is a property of the set, not of the selector.

4. The geometry–levels decomposition `[FINAL]`

Theorem 3.2 says error lives in the wobble. Where does the wobble live? Comparing model-predicted and reference image-energy argmax on the locked panel:

Model	exact saddle location	within ±1 image	along-path wobble (RMS)
chgnet	24/28 (86%)	26/28 (93%)	~131 meV
mace-mp-small	23/26 (88%)	23/26 (89%)	~103 meV
mace-mpa-0-medium	23/28 (82%)	25/28 (89%)	~110 meV

The models misprice the *level* of the landscape while correctly locating its *extrema*. This is the decomposition the whole program turns on: geometry is the model's skill; levels are its failure. The published softening record attributes level failure to near-equilibrium training bias (Deng et al.); our measurement adds that the *geometric* layer survives that bias.

5. Method: model-guided sparse DFT (frozen protocol)

Theorem 3.4 prices exactness at two anchors — provided they are the *true* extrema. §4 says the model supplies them 82–93% of the time. The protocol follows:

1. **Guide.** Each model evaluates the path's input images (single-point energies; no relaxation — the model only guides) and proposes predicted min/max image indices.
2. **Anchors.** A free DFT engine (GPAW, PBE [enkovaara2010gpaw; mortensen2005gpaw; perdue1996pbe]) evaluates the reference-profile positions `{min, max ± 1}` plus both endpoints; short paths (≤6 images) widen to ±2, declared.
3. **Measure.** Sparse barrier = max − min over anchor energies, scored against the locked reference barrier. **WIN: MAE ≤ 40 meV per model. Strong WIN: MAE ≤ 15 meV.**
4. **Honesty conditions.** Anchors beyond the frozen sets void the cost claim; GPAW-vs-VASP convention differences (functional, pseudopotential, k-points; [grimme2006d2; kresse1996vasp]) are reported as their own line item; any GPAW failure fails that path, recorded without imputation.

Simulation on the reference profile `[FINAL]`: across all 82 model-path pairs, MAE 1.2–9.4 meV at ≈7 anchors/path (±2: exact) — the same 40 meV gate the raw models miss by 3.4–6×, at ≈10× lower DFT cost.

6. Results (preregistered analysis plan)

- **T1 — convention offset**: GPAW vs VASP reference energy per anchor (mean, max |Δ|), per path class. `[PENDING]`
- **T2 — sparse MAE per model** with WIN/strong-WIN verdicts and per-family no-harm checks. `[PENDING]`
- **T3 — cost**: median anchors/path vs the frozen full-NEB protocol. `[PENDING]`
- **F1 — the boundary figure**: raw error vs wobble, sparse error vs wobble, Theorem 3.2 overlaid. `[PENDING]`

7. The process: distilling physical laws from error structure

The measurements in this memorandum were produced by a pipeline that treats *process* as a first-class scientific object: campaign manifests are content-addressed before execution; panels are SHA-locked; runners validate hashes fail-closed; measurement rows are RFC 8785-canonicalized and hash-chained;

ingestion into the claims registry is derivation, not editing — status changes require bundle-hash changes, so no verdict can be laundered by hand; and the formal layer certifies the registry's outcomes in Lean (`z1_gate_refuted`, `z3_gate_refuted` — the machine-checked record of these same failures). Abstention is a first-class output: the registry of this program contains refuted claims and an abstained campaign, deliberately un-deleted. We claim this process — preregistration + hash-chained evidence + machine-checked claims + abstention — is itself a method for distilling physical laws from error structure, and the correction boundary of §3 is its first law.

8. Related work

Softening and underbinding. Deng et al., npj Comput. Mater. (2025), DOI 10.1038/s41524-024-01500-6; Bheemaguli, Xiao & Sai Gautam, Digital Discovery 5 (2026) 1809–1819, DOI 10.1039/D5DD00534E. **Barrier-specific fine-tuning (the training route we exclude).** Lian et al., J. Mater. Chem. A 13 (2025), DOI 10.1039/D5TA05355B — effective at ~3k labels; our constraint is zero training. **Δ-learning and active residual models.** Christiansen & Hammer, J. Chem. Phys. 162, 184701 (2025), DOI 10.1063/5.0268264; Pitfield et al., Phys. Chem. Chem. Phys. 28, 912–926 (2026), DOI 10.1039/D5CP04302F. **Foundation models.** MACE-MP-0: Batatia et al., arXiv:2401.00096; CHGNet: Deng et al., Nat. Mach. Intell. 5, 1031–1041 (2023), DOI 10.1038/s42256-023-00716-3; DPA-1/DPA-2: Zhang et al., npj Comput. Mater. (2024), DOIs 10.1038/s41524-024-01278-7, 10.1038/s41524-024-01493-2; MatterSim: Yang et al., arXiv:2405.04967. **Benchmarks.** LiTraj nebDFT2k, DOI 10.1038/s41524-025-01571-z; CatBench BM (GAME-Net lineage), Zenodo DOI 10.5281/zenodo.17157086, npj/Nat. Comput. Sci. DOI 10.1038/s43588-023-00437-y. **Method theory.** Sloppy-model/error-geometry foundations: Brown & Sethna, Phys. Rev. E 68, 021904 (2003). CI-NEB: Henkelman & Jónsson, J. Chem. Phys. 113, 9978 (2000). Conformal abstention: Vovk, Gammerman & Shafer (2005); Angelopoulos & Bates, arXiv:2107.07511. **Verification.** Lean 4: de Moura & Ullrich, CADE-28 (2021). Preregistration practice: Nosek et al., PNAS 115, 2600–2606 (2018). **Quantum engines.** GPAW: Enkovaara et al., J. Comput. Chem. 31, 2474 (2010); Mortensen et al., Phys. Rev. B 71, 035109 (2005). VASP: Kresse & Furthmüller, Phys. Rev. B 54, 11169 (1996). PBE: Perdew, Burke & Ernzerhof, Phys. Rev. Lett. 77, 3865 (1996). Grimme D2: J. Comput. Chem. 27, 1787 (2006).

9. Epistemic-status register

Claim	Status	Receipt
Four models fail the 40 meV barrier gate, 135–243 meV	FINAL	Round-4 locks + artifacts, PRs #26–#31
Precision-independence of the failure	FINAL	float64 chain, PR #32, #36
All cheap corrections fail; pilot abstains universally	FINAL	R5 record, PRs #54, #55

Claim	Status	Receipt
Routing oracle ceiling 70 meV	FINAL	R4 analysis (this repo)
Boundary theorems (3.1–3.4, Corollary 3.5)	MACHINE-CHECKED	`CorrectionBoundary.lean`, PR #56
Geometry accuracy 82–86% exact / 89–93% ±1	FINAL	R4 artifacts
Sparse protocol MAE 1.2–9.4 meV on reference profile	FINAL (simulation)	analysis of locked panel
Sparse pilot physical MAE, WIN verdicts, cost	PENDING	campaign per §5
GPAW-vs-VASP convention offset	PENDING	T1

10. Limitations and honesty conditions

Panel paths are short (5–10 images); the sparse advantage grows with denser paths and is not claimed at long-path scale. The ±2 short-path fallback is declared, not tuned. Simulation (§5) is exact arithmetic on reference data, not physical measurement; the campaign decides, and if geometry guidance degrades on denser paths that failure is reported with the same prominence as a WIN. The abstention record of §2.3 stands; nothing here re-opens the corrections already refuted. All panel references are published DFT, not experiment; nothing here is error-against-experiment.

References

[Resolved in §8 with DOIs; additional: Brown & Sethna PRE 68, 021904 (2003); Henkelman & Jónsson JCP 113, 9978 (2000); Vovk et al. (2005); Angelopoulos & Bates arXiv:2107.07511; de Moura & Ullrich, CADE-28 (2021); Nosek et al., PNAS 115 (2018); Enkovaara et al., JCC 31 (2010); Mortensen et al., PRB 71 (2005); Kresse & Furthmüller PRB 54 (1996); Perdew, Burke & Ernzerhof PRL 77 (1996); Grimme, JCC 27 (2006).]

Receipts: locked panels (`192fe54a…`, `4099f4fc…`); Round-4/5 artifacts `gs://shed-489901-atlas-outputs/`; Lean module PR #56; preregistrations `docs/plans/2026-07-20-sparse-dft-pilot-preregistration.md`, `2026-07-20-round5-z1-correction-preregistration.md`, `2026-07-20-correction-boundary-theorems.md`; claims registry entries for this campaign remain `unsupported` until ingestion.